

Monotonicity of the Ratios

1 Definitions

Definition 1 (Polyomino). A *polyomino* is a finite set $P \subset \mathbb{Z}^2$ of unit square cells that is connected by edge adjacency. Its *area* (or size) is $|P|$; a polyomino of area n is called an n -omino.

Definition 2 (Fixed and free polyominoes). Two polyominoes are *fixed-equivalent* if they differ by a translation. Let $A(n)$ be the number of fixed n -ominoes.

If rotations and reflections are also identified, one obtains *free polyominoes*. Since every free class contains at most eight fixed classes, fixed and free polyominoes have the same exponential growth constant. We focus on $A(n)$.

Definition 3 (Klarner's constant). The *Klarner constant* is

$$\lambda = \lim_{n \rightarrow \infty} A(n)^{1/n} = \lim_{n \rightarrow \infty} \frac{A(n+1)}{A(n)}. \quad (1)$$

2 Current best result

The ratios $A(n+1)/A(n)$ are known to converge to λ . All available exact enumeration data are consistent with these ratios being strictly increasing, but eventual monotonicity has not been proved.

3 Problem formulation

Prove that, for all sufficiently large n ,

$$\frac{A(n+1)}{A(n)} < \frac{A(n+2)}{A(n+1)}. \quad (2)$$